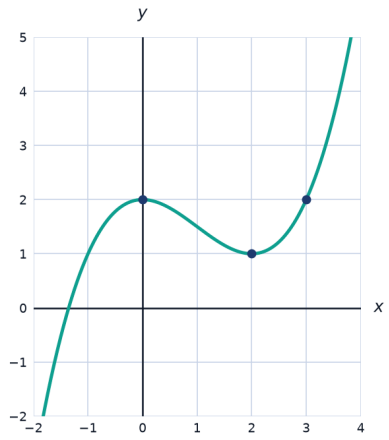
Algebra: Partial Fractions & Division

Section 1: Polynomial Division & Improper Fractions

1 Divide $2x^3 - 3x^2 + 4x - 5$ by $(x - 2)$ to find the quotient and remainder.



a. Quotient $= 2x^2 + x + 6$, Remainder $= 7$

b. Quotient $= 2x^2 - 7x + 18$, Remainder $= -41$

c. Quotient $= 2x^2 + x + 6$, Remainder $= -7$

d. Quotient $= 2x^2 + 7x + 18$, Remainder $= 31$

2 Express the improper rational function $\dfrac{x^2 + 5x + 7}{x + 2}$ in the form $Ax + B + \dfrac{C}{x + 2}$.

a. $x + 3 + \dfrac{1}{x + 2}$

b. $x + 3 + \dfrac{7}{x + 2}$

c. $x + 7 - \dfrac{7}{x + 2}$

d. $x + 2 + \dfrac{3}{x + 2}$

3 When $P(x) = x^4 + ax^2 + bx - 4$ is divided by $(x - 1)$, the remainder is 2; when divided by $(x + 2)$, the remainder is -8. Find a and b.

a. $a = 1, b = 4$

b. $a = 2, b = 3$

c. $a = -1, b = 6$

d. $a = 3, b = 2$

Section 2: Partial Fractions with Linear & Repeated Factors

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4 Express $\dfrac{7x-1}{(x-1)(x+2)}$ in partial fractions.

a. $\dfrac{2}{x-1}+\dfrac{5}{x+2}$

b. $\dfrac{5}{x-1}+\dfrac{2}{x+2}$

c. $\dfrac{6}{x-1}+\dfrac{1}{x+2}$

d. $\dfrac{7}{x-1}-\dfrac{1}{x+2}$

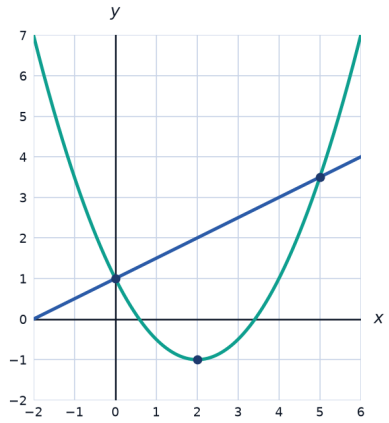
5 What is the correct partial fraction decomposition template for $\dfrac{3x^2+1}{(x-2)^2(x+3)}$?

a. $\dfrac{A}{x-2}+\dfrac{B}{(x-2)^2}+\dfrac{C}{x+3}$

b. $\dfrac{A}{(x-2)^2}+\dfrac{B}{x+3}$

c. $\dfrac{Ax+B}{(x-2)^2}+\dfrac{C}{x+3}$

d. $\dfrac{A}{x-2}+\dfrac{B}{x+3}$



6 In the decomposition $\dfrac{4x}{(x-1)^2}=\dfrac{A}{x-1}+\dfrac{B}{(x-1)^2}$, find A and B.

a. $A=4, B=4$

b. $A=4, B=0$

c. $A=0, B=4$

d. $A=2, B=4$

Section 3: Irreducible Quadratic Factors

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7 What is the correct partial fraction template for $\frac{2x+5}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$

b. $\frac{A}{x-1}+\frac{B}{x^2+4}$

c. $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

d. $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

8 In $\frac{7x^2+5x+13}{(x-1)(x^2+4)}=\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$, find the value of A.

a. 5

b. 7

c. 2

d. 3

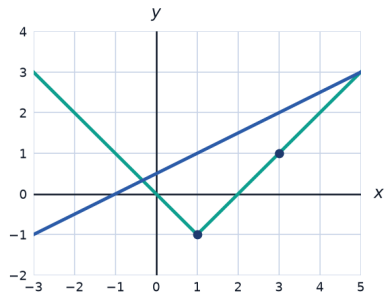
9 For the decomposition in the previous question ($A=5$), find B and C.

a. $B=2, C=7$

b. $B=2, C=-7$

c. $B=7, C=2$

d. $B=-2, C=7$



Section 4: Binomial Series with Partial Fractions

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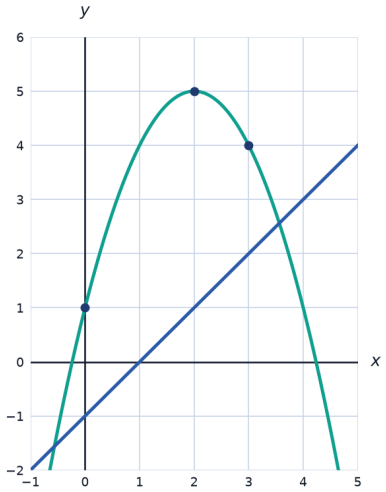
10 To expand $f(x) = \dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of x for small $|x|$, which partial fraction form should be expanded?

a. $-\dfrac{2}{1-x} + \dfrac{5/2}{1+x/2}$

b. $\dfrac{2}{x-1} + \dfrac{5}{x+2}$

c. $\dfrac{2}{1-x} + \dfrac{5}{1+x/2}$

d. $-\dfrac{2}{1-x} + \dfrac{5}{1+x}$



11 Find the constant term in the expansion of $\dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $\dfrac{1}{2}$

b. $-\dfrac{1}{2}$

c. -2

d. 5

12 State the radius of convergence for the expansion of $\dfrac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $|x| < 1$

b. $|x| < 2$

c. $|x| < 1/2$

d. $|x| < 7$

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